Stability vs Adaptability in Artificial General Intelligence

A NARS-Based Experimental Study of Belief Formation Without Large-Scale Training

Individual Research Project — Temple University, Philadelphia

The problem with modern AI

- Systems like GPT need billions of training examples before they can do anything useful.
- Training takes weeks on thousands of computers.
- Once training ends, learning stops completely.
- Nobody can look inside and see what the system actually believes.

This project asks: is all of that really necessary for intelligence?

What is NARS?

NARS stands for **Non-Axiomatic Reasoning System**. It forms beliefs immediately from any single input.

Every belief is stored as exactly two numbers:

- **Frequency (f):** fraction of observations that confirmed the belief (range 0.0 to 1.0).
- **Confidence (c):** how much total evidence exists, using $c = k / (k + 1)$ where k is total observations.

No pre-training needed. No large dataset required. Learns and reasons at the same time.

Research question

> *Can a system build meaningful beliefs from scratch — with no pre-training and minimal data — and still maintain the right balance between holding onto what it learned and updating when it should?*

Three hypotheses

H1

Beliefs from zero knowledge

NARS can form stable, meaningful beliefs starting from zero knowledge, using just a few observations.

H2

Resistance and revision

Once formed, beliefs resist small contradictions but revise appropriately when contradictory evidence is strong enough.

H3

Improvement through experience

Belief quality — both frequency and confidence — improves over time through experience alone, with no pre-training phase required.

Experimental design — three stages

Stage 1

Cold Start (Week 2)

System begins with zero knowledge. Fed uncertain, incomplete, and contradictory inputs.

Measures whether any belief forms at all.

Stage 2

Reinforcement (Week 3)

Confirming observations repeated in batches of 5, 10, and 20.

Measures how quickly belief stabilises and when confidence stops growing.

Stage 3

Stress Test (Week 4)

Contradictions injected into stable beliefs at light, moderate, and heavy levels.

Measures resistance and whether revision is proportional.

Stage 1 — Cold start belief formation

- System starts knowing absolutely nothing.
- After just one observation, a belief immediately forms.
- Frequency reflects the ratio of confirming to total observations.
- Confidence grows with every input via **c = k / (k+1)**.
- Even uncertain and contradictory inputs produce measurable beliefs from the very first step.

Stage 1 — Cold start belief formation (zero prior knowledge)
"Birds can fly" — step by step

Observation	Type	Confirms	Contradicts	Total (k)	Frequency (f)	Confidence (c)
Sparrow flies	Confirms	1	0	1	1.000	0.500
Eagle soars	Confirms	2	0	2	1.000	0.667
Robin flies	Confirms	3	0	3	1.000	0.750
Pigeon takes off	Confirms	4	0	4	1.000	0.800
Hawk circles	Confirms	5	0	5	1.000	0.833
Penguin walks	Contradicts	5	1	6	0.833	0.857
Ostrich runs	Contradicts	5	2	7	0.714	0.875
Swallow dives	Confirms	6	2	8	0.750	0.889
Kiwi hides	Contradicts	6	3	9	0.667	0.900
Crane lifts off	Confirms	7	3	10	0.700	0.909

Stage 1B — Reasoning from incomplete information

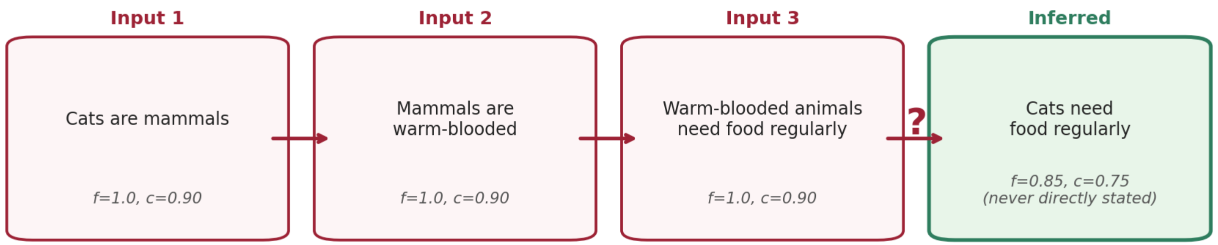
The system was given three separate facts:

- Cats are mammals.
- Mammals are warm-blooded.
- Warm-blooded animals need food regularly.

It was never directly told that cats need food regularly. NARS derived this on its own by chaining the three facts together.

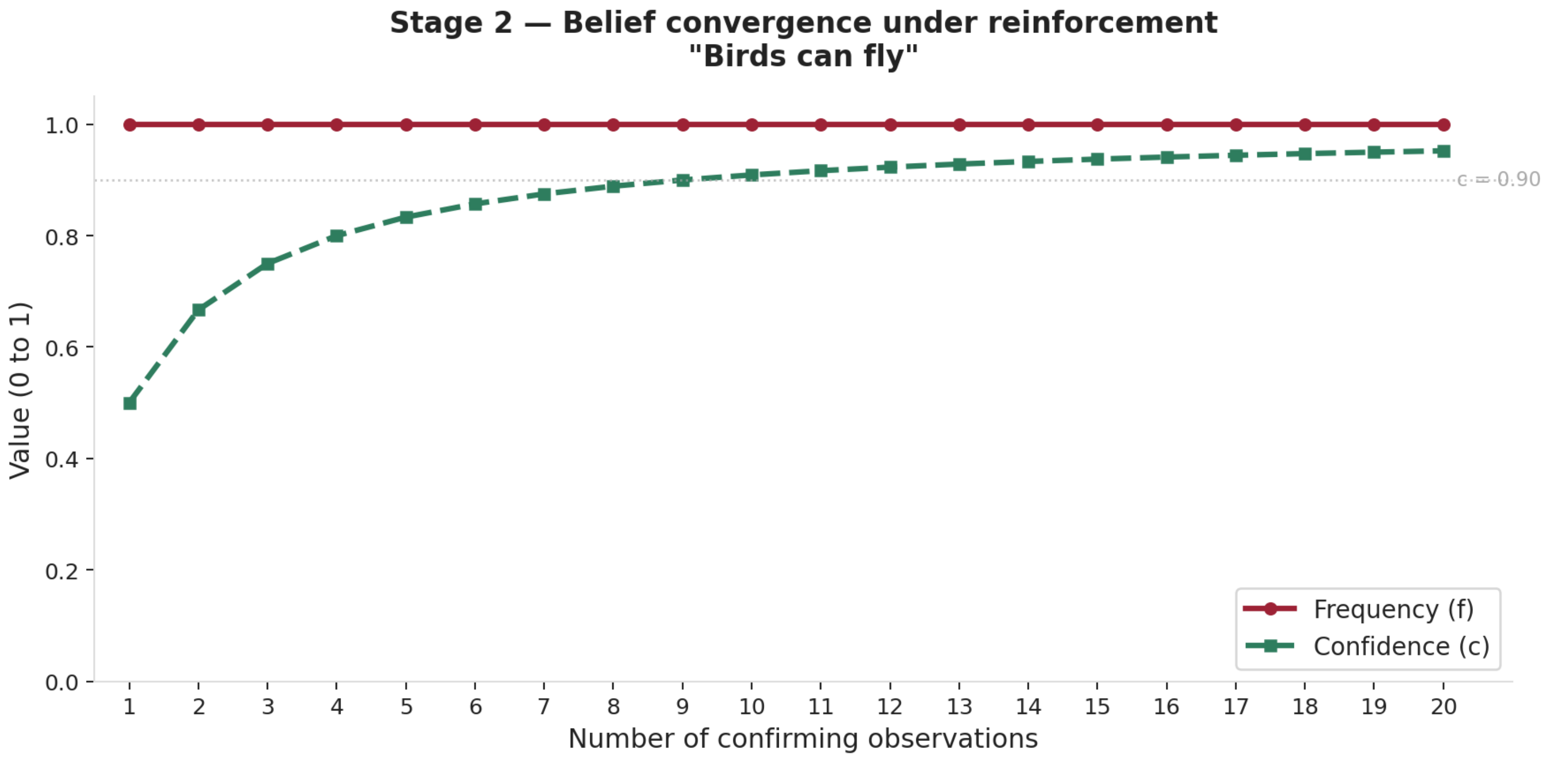
This is deductive inference — reasoning beyond what you were explicitly taught.

Stage 1B — Incomplete input inference chain
(System infers conclusion it was never directly told)



Stage 2 — Belief convergence under reinforcement

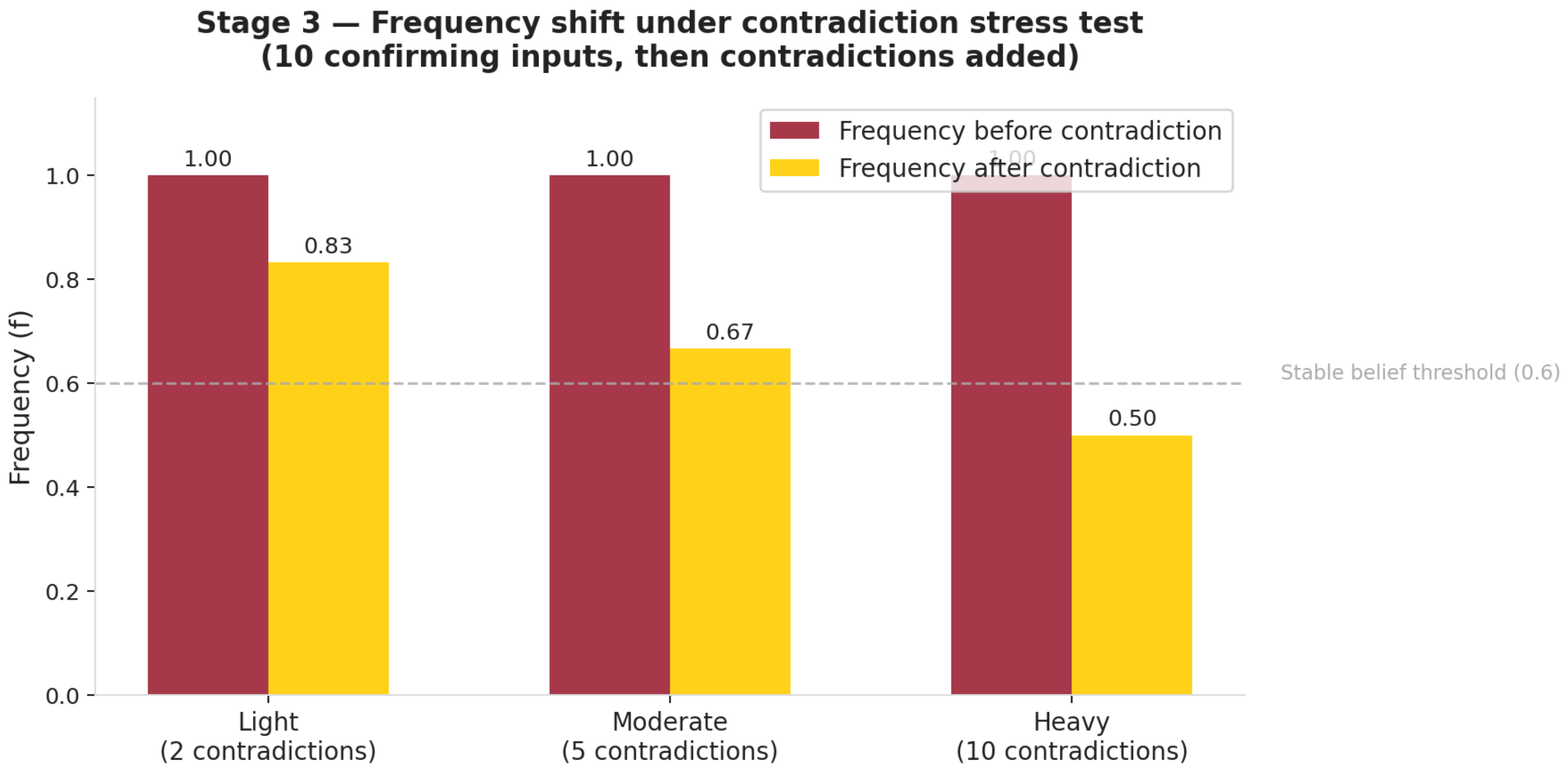
- As confirming observations increase, frequency stays high and stable near 1.0.
- Confidence rises steadily but follows a diminishing-returns curve — each new observation adds less than the last.
- This is the **c = k / (k+1)** formula in action.
- After ~10 observations the belief becomes strongly stable and difficult to shift.



Stage 3 — Resistance to contradiction

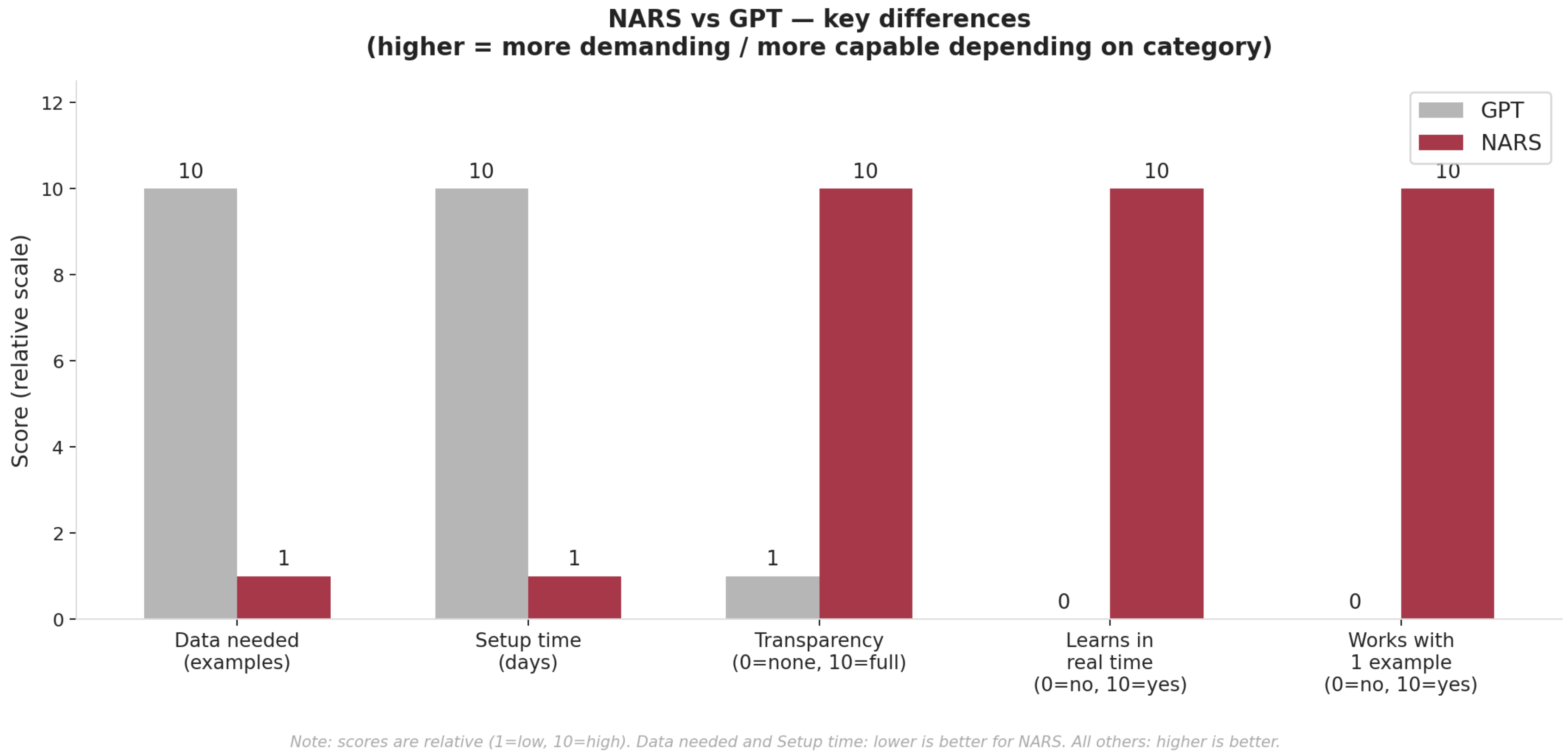
- **Light (2 contradictions after 10 confirms):** frequency drops slightly, belief holds above 0.6.
- **Moderate (5 contradictions after 10 confirms):** frequency drops to ~0.67, still leans true.
- **Heavy (equal confirms and contradictions):** frequency reaches 0.5, belief becomes genuinely uncertain.

A belief built on more evidence needs proportionally more contradictions to overturn.



NARS vs GPT — key differences

- **Data:** NARS learns from 1 example; GPT needs billions.
- **Learning:** NARS learns continuously in real time; GPT learns only in a separate training phase.
- **Transparency:** NARS beliefs are fully visible as (f, c); GPT hides them inside billions of weights.
- **Setup:** NARS needs zero setup time; GPT needs weeks of training.
- **Contradictions:** NARS drops frequency immediately and visibly; GPT silently averages them out.



Conclusion

This project demonstrates three things:

1. A reasoning system can build meaningful beliefs from scratch with no pre-training — a handful of observations is enough.
2. Those beliefs become resistant to contradiction as evidence accumulates, yet still revise proportionally when contradictions are strong enough.
3. Stability and adaptability are not opposing forces — they are complementary properties of the same intelligent system.

General intelligence does not require massive training data — it requires a well-designed reasoning framework and experience.